A REPORT

ON

SMART EXAM SCHEDULER

By

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CERTIFICATE

This is to certify that the project entitled "**SMART EXAM SCHEDULER**" has been successfully completed by **Ch. Vedaswi (AP24110010964), Sd. Ruda Kausar(AP24110011036), N. Sujitha (AP24110011030)** as a required academic component of the course *CSE 206: Coding Skills* during Semester 4 of Academic Year 2025-26 in the Department of Computer Science and Engineering at SRM University, AP. This work was executed under the guidance of the undersigned and is deemed to have successfully met all course requirements as of April **2026.**

Abstract

The Smart Journey Planner is a C++ based application designed to help users find the most efficient route between cities. It uses graph-based algorithms to calculate the shortest distance and minimum travel cost. The system allows users to input city routes, view connections, and compute optimized paths using Dijkstra's Algorithm and Dynamic Programming techniques.

Introduction

With increasing travel needs, finding the best route between cities is essential. This project aims to simplify route planning by providing a system that calculates:

- Shortest distance between cities
- Minimum travel cost

The application uses Data Structures and Algorithms (DSA) concepts like graphs to represent city connections and efficient algorithms to compute optimal paths.

Objectives

- To design a route planning system using graphs
- To implement shortest path calculation using Dijkstra Algorithm
- To calculate minimum travel cost using Dynamic Programming
- To provide a simple user interface for interaction

Technologies Used

- **Programming Language:** C++
- **IDE:** Visual Studio Code
- **Concepts Used:**
  - Graphs
  - Greedy Algorithm (Dijkstra)
  - Dynamic Programming

Algorithm Explanation

Dijkstra Algorithm

- Finds shortest distance from a source city to all other cities
- Works by selecting the minimum distance node repeatedly
- Time Complexity: **$O(n^2)$**

Dynamic Programming (Cost Calculation)

- Calculates minimum travel cost
- Updates cost values iteratively
- Efficient for multi-step route optimization

Sample Input & Output

```
Enter number of cities: 4

1.Add Route
2.Show Routes
3.Shortest Distance
4.Minimum Cost
5.Exit
1
Enter source city number: 0
Enter destination city number: 1
Enter distance: 10
Enter cost: 5

1.Add Route
2.Show Routes
3.Shortest Distance
4.Minimum Cost
5.Exit
1
Enter source city number: 1
Enter destination city number: 2
Enter distance: 20
Enter cost: 4

1.Add Route
2.Show Routes
```

```
3.Shortest Distance
4.Minimum Cost
5.Exit
2
Distance Matrix
0 10 0 0
0 0 20 0
0 0 0 0
0 0 0 0

1.Add Route
2.Show Routes
3.Shortest Distance
4.Minimum Cost
5.Exit
3
Enter source city: 1
Shortest Distances:
City 0 : 2147483647
City 1 : 0
City 2 : 20
City 3 : 2147483647
```

```
1.Add Route
2.Show Routes
3.Shortest Distance
4.Minimum Cost
5.Exit
4
Enter source city: 1
Minimum Cost to cities:
City 0 : 2147483647
City 1 : -2147483644
City 2 : -2147483640
City 3 : 2147483647

1.Add Route
2.Show Routes
3.Shortest Distance
4.Minimum Cost
5.Exit
5


=== Code Execution Successful ===
```

Advantages

- Easy to use
- Efficient route calculation
- Uses optimized algorithms
- Can be extended for real-world applications

Limitations

- Limited to 10 cities (fixed array size)
- No graphical interface
- No real-time data integration

Conclusion

The Smart Journey Planner successfully demonstrates the use of graph algorithms in solving real-world problems like route optimization. By implementing Dijkstra's Algorithm and Dynamic Programming, the system efficiently computes shortest paths and minimum costs. This project provides a strong foundation for building advanced navigation systems.